

IT Service Desk Ticket Assistant

An AI-first service desk: employees get help through chat or a ticket form, the bot resolves what it safely can from the knowledge base, and the support team works every escalation with full context, priority alerts and performance reporting.

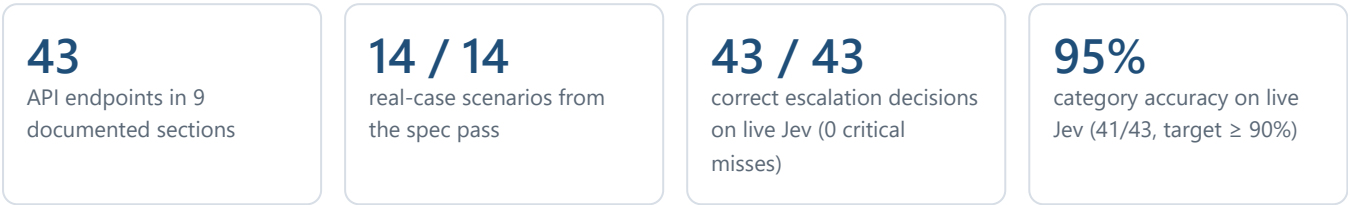
Date	28 September 2026
Build	servicedesk_jev · Phase 2 · FastAPI + Streamlit + SQLite · TypeSafe Jev (pinned jev-1.13.0)
Status	All Phase 2 items built · 29/29 automated checks pass · spec scenarios 14/14
Next phase	Phase 5 — email channel

Contents

1 Executive summary	10 Workflow E — working a ticket (support)
2 Purpose and problem	11 Ticket lifecycle
3 The solution at a glance	12 Supervisor insight: dashboards and reports
4 Architecture	13 Safety, security and conflict rules
5 People, roles and sign-in	14 Quality: tests and evaluation
6 Workflow A — employee chat, end to end	15 API reference (every endpoint)
7 Workflow B — the ticket form	16 Function reference
8 Workflow C — My tickets and notifications	17 Running the system
9 Workflow D — alerts and escalation	18 Limitations and roadmap

1 Executive summary

The IT Service Desk Ticket Assistant gives every employee one place to get IT help and gives the support team one place to work. A bot handles the first line: it understands the issue, checks it for safety and security risk, tries at most two fixes taken word-for-word from the knowledge base, and hands anything it can't safely solve to the right team with the whole conversation attached. People never repeat themselves, critical issues can't sit unnoticed, and every bot decision is visible and correctable.



What Phase 2 delivered

Area	Delivered
2A Foundations	Sign-in with roles, locked sessions and lockout; one API as the only writer; a ticket state machine that rejects illegal jumps; ownership claims and version locking; accuracy fixes with a locked evaluation baseline.
2B Employee	Sign-in page and "Can't sign in?" recovery; chat on the API; a guided ticket form with review, submit and "fix now / just log"; My tickets with reply, reopen, confirm and cancel; duplicate guard; in-app notifications.
2C Support	Priority queue; ServiceNow-style ticket form with bot panel, suggested KB, checklist, comments and work notes, activity timeline and a "People on this ticket" panel; P1/P2 alerts with re-alert and supervisor escalation; charts C1–C12; agent performance; shift summary and checklist; corrections log.

Design rule that makes it safe: Jev (the AI model) only answers typed questions with probabilities. Code owns priority, routing, verification, attempts and state. Replies are knowledge-base text or fixed templates, so the bot cannot invent a fix.

2 Purpose and problem

The service desk exists to get employees working again quickly and safely. Before this build, each group had a different pain:

Who	Pain before
Employees	Repeat details to every agent; wait in queues for simple fixes; can't see status.
L1 agents	Overloaded by password, VPN and printer repeats; tickets arrive without context.
Support leads	Critical issues sit unnoticed; no view of SLA risk or workload.
Organisation	Can't prove automation helps; bot mistakes are invisible and never corrected.
Previous build	No logins; bot decisions only in a trace panel; no alerts, metrics, ownership lock or checklist; two processes writing one database.

Goals for Phase 2

- Every user signs in; each person sees only what their role allows, and identity comes only from the sign-in token.
- A critical ticket always reaches a human within one poll cycle (≤ 10 seconds) and keeps alerting until acknowledged.
- Agents get the whole story — conversation, bot reasoning, steps already tried — and a checklist that makes resolution consistent.
- Leads can see volume, SLA, bot effectiveness and safety at a glance, and every bot mistake a person corrects becomes improvement data.

3 The solution at a glance

Side	What they get
Employee	Sign in → Chat or Ticket form → My tickets (status, thread, reply, reopen) → in-app notifications.
Support agent	Sign in → queue with P1s on top → priority alerts → ticket form with bot panel, suggested KB, checklist, comments/work notes, activity timeline and people panel.
Supervisor	Everything an agent has, across all queues, plus charts, agent performance, shift summary and the corrections log.
Bot	Triage, knowledge-base-grounded attempts (max 2), verified tool actions, safe escalation with a full handoff package.

Design principles

1. **One pipeline, many doors.** Chat and form (and email in Phase 5) share one main agent and one set of policy gates.
2. **Jev decides, code enforces.** Jev returns typed judgments; code owns priority, gates, verification, attempts and state.
3. **No generated fixes.** Replies are templates or knowledge-base text.
4. **Policy beats confidence.** Safety and security flags override category and priority.
5. **Trust comes from the channel, not the message.** A signed-in session is verification level V1; sensitive actions need V2 or higher.
6. **One writer, one state machine, append-only events.**
7. **Nothing is deleted.** Tickets are cancelled or closed, never removed.

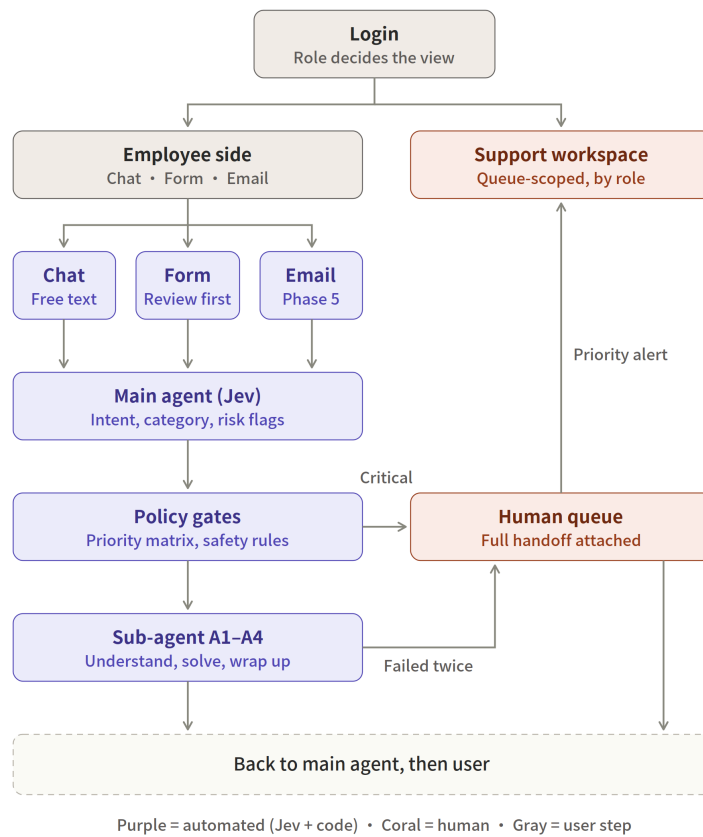


Figure 1. System overview: every channel enters one pipeline of triage, policy gates and sub-agents.

4 Architecture

The browser app (Streamlit) never touches the database. It calls the FastAPI service with the signed-in user's token, and the API is the only component that writes. This removed the old risk of two processes writing one database and makes it impossible to act as another user by editing a request.

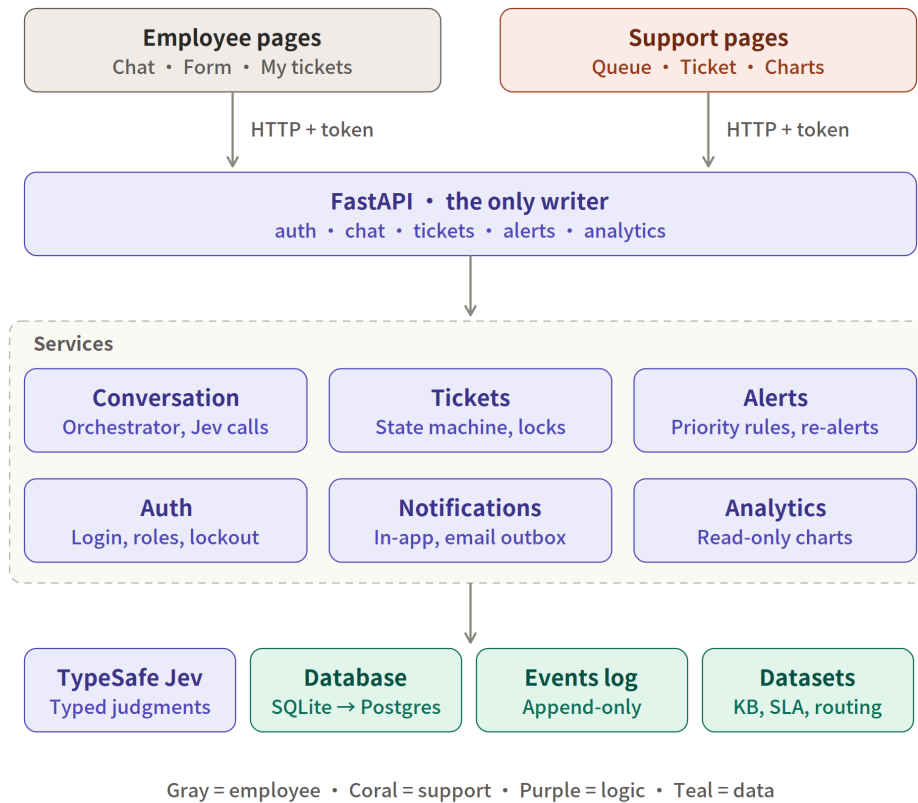


Figure 2. Layered architecture: UI client → API (only writer) → services → SQLite, with Jev behind one gateway.

Services and their boundaries

Service	Owns	Never does
ConversationService	Chat and form turns, Jev calls, knowledge-base attempts	Change status directly (it asks TicketService)
TicketService	Create, transition, claim, versioned update, resolve	Talk to Jev
AlertService	Alert rules, re-alerts, supervisor escalation, acknowledgements	Send messages itself
NotificationService	In-app notifications (email outbox in Phase 5)	Decide who is alerted
AnalyticsService	Read-only aggregates for charts and reports	Write anything
AuthService	Sign-in, sessions, roles, lockout, recovery	Grant V2+ verification (the identity tool does)

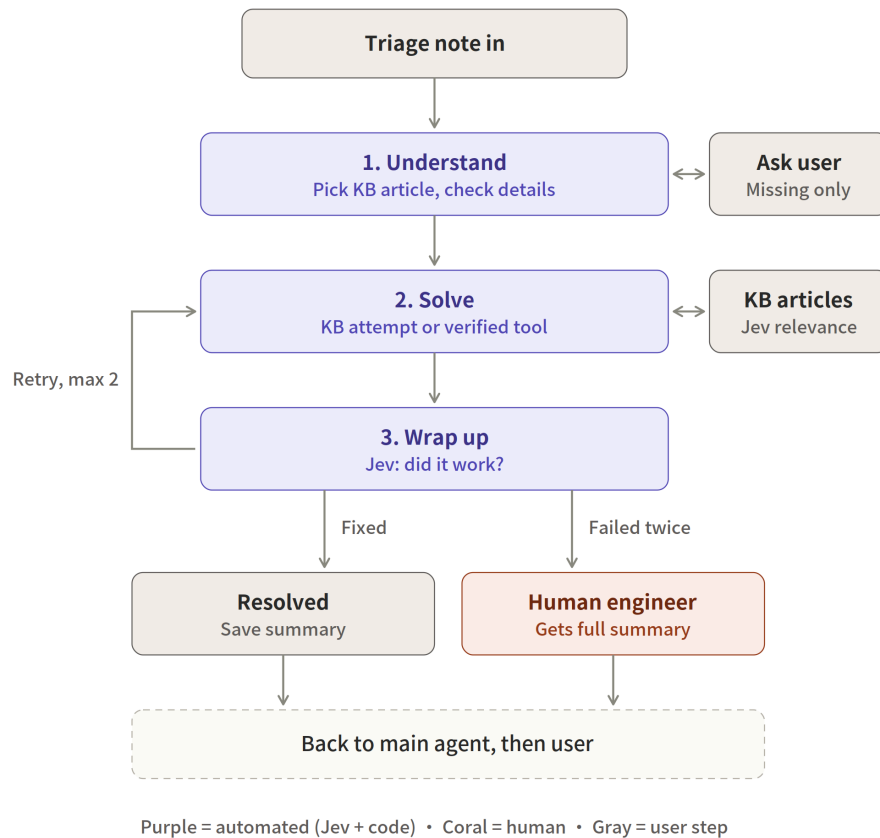


Figure 3. Each sub-agent works in three roles: Understand (Jev scores KB articles and required fields), Solve (max two attempts) and Wrap up.

Technology

Layer	Choice	Why
AI model	TypeSafe Jev (System One), pinned to jev-1.13.0	Typed answers with probabilities, so code can gate on confidence; no free text to hallucinate.
API	FastAPI + Pydantic	Single writer, role checks on every route, self-documenting at /docs.
UI	Streamlit 1.64 + Plotly	Role-based pages, 10-second alert polling, dialogs and toasts, interactive charts.
Data	SQLite in WAL mode	Concurrent reads with one writer; append-only event log; migrations from Phase 1.
Security	bcrypt, JWT + hashed session rows	Revocable sessions with idle and absolute expiry; lockout after 5 failures.

5 People, roles and sign-in

Everyone signs in the same way: **employee ID or work email** and a password. IT staff are employees too, so they have ordinary employee IDs; the role (employee, agent or supervisor) comes from the directory, not from the ID.

Figure 4. The single sign-in page for everyone, with “Can’t sign in?” recovery underneath.

Role	Employee IDs	Can use
Employee	EMP1001 – EMP1050	Chat, Ticket form, My tickets — own data only
Agent	EMP2002 – EMP2015	Own queues + every P1, alerts, ticket form, bot answers, charts (own scope), own performance, shift
Supervisor	EMP2001	All queues, all charts, everyone's performance, corrections log, priority override

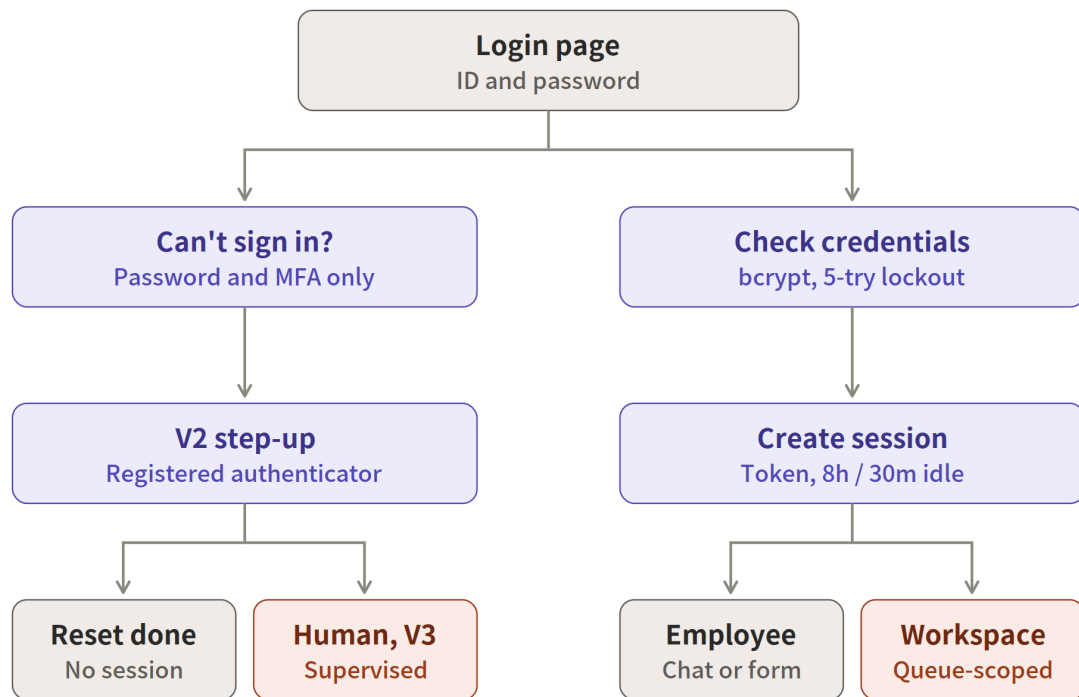
IT staff directory

Employee ID	Name	Job title	Queues
EMP2001	Lakshmi Narayanan	Service Desk Supervisor	All queues
EMP2002	Priya Sharma	IT Support Engineer	Application Access
EMP2003	Arjun Menon	IT Support Engineer	Collaboration Support
EMP2004	Meera Pillai	Desktop Support Engineer	End User Hardware
EMP2005	Rahul Verma	Identity Security Analyst	Identity Security
EMP2006	Kavya Reddy	Identity Support Engineer	Identity Support
EMP2007	Vikram Singh	Messaging Engineer	Messaging Support

Employee ID	Name	Job title	Queues
EMP2008	Divya Nair	Mobile Device Engineer	Mobile Device Management
EMP2009	Karthik Rao	Network Engineer	Network Operations
EMP2010	Ananya Iyer	Network Engineer	Network Remote Access
EMP2011	Rohan Gupta	Security Operations Analyst	Security Operations
EMP2012	Sneha Kulkarni	Service Desk Duty Manager	Service Desk Duty Manager
EMP2013	Aditya Joshi	Software Asset Analyst	Software Asset Management
EMP2014	Nisha Kapoor	Workplace Support Engineer	Workplace/Print Support
EMP2015	Samuel Thomas	Network Engineer	Network Remote Access, Network Operations

How sign-in is protected

Control	Behaviour
Passwords	Stored as bcrypt hashes; the demo password lives in .env, never in code.
Lockout	5 wrong passwords lock the account for 15 minutes; every attempt is audited.
Sessions	Signed JWT plus a hashed session row: 8-hour absolute limit, 30-minute idle limit, revoked on sign-out.
Identity	Taken only from the token. The API ignores any user ID sent in a request body.
Roles	Checked on every route; an employee opening a support page gets 403 role_denied, which is audited.
Recovery	"Can't sign in?" sends a simulated push to the registered authenticator (V2) or opens a supervised V3 ticket. It never signs anyone in.



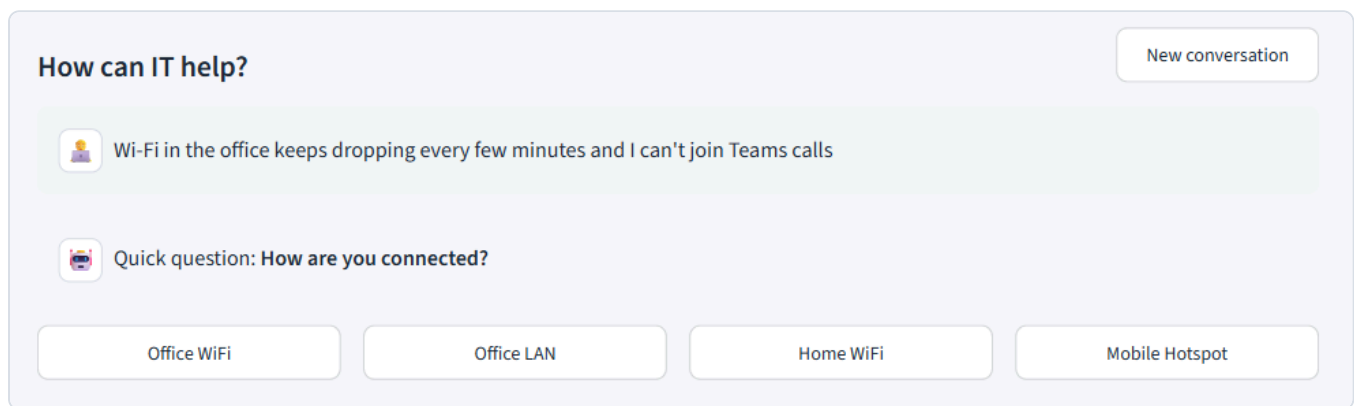
Every attempt is logged • login gives V1 only; resets still need V2

Figure 5. Sign-in and recovery flow.

6 Workflow A — employee chat, end to end

This walk-through uses real screens. An employee reports unstable office Wi-Fi; the bot asks what it needs, tries two fixes from the knowledge base, and escalates with a full handoff when neither works.

- 1 **Describe the issue.** Jev triages the message in one call: intent, category, impact, urgency and eight safety/policy flags.
- 2 **Answer quick questions.** Only the fields the category needs and the message didn't already answer (connection type, scope).
- 3 **Attempt 1 and attempt 2.** Steps are copied from the best-scoring KB article (here KB-022). After each, the bot asks whether it worked.
- 4 **Escalate with context.** After two failures the ticket goes to the routed queue with priority, SLA and the whole conversation.



The screenshot shows a chat window titled "How can IT help?". In the top right corner, there is a button labeled "New conversation". The chat history shows a user message: "Wi-Fi in the office keeps dropping every few minutes and I can't join Teams calls". Below this, a bot response is shown: "Quick question: How are you connected?". At the bottom of the chat window, there are four buttons for quick replies: "Office WiFi", "Office LAN", "Home WiFi", and "Mobile Hotspot".

Figure 6. The employee describes the issue; the bot asks one targeted question with quick-reply buttons.

How can IT help?

New conversation

 Wi-Fi in the office keeps dropping every few minutes and I can't join Teams calls

 Quick question: **How are you connected?**

 Office WiFi

 Quick question: **Scope of issue** (e.g. Which sites/apps fail?)

 Teams calls and SharePoint; other sites load slowly too

 **Attempt 1 of 2** · from *KB-022 · No Internet on Office WiFi*
👉 Basic reset (forget/rejoin network).
Then try again — did that fix it?

Yes, it's fixed

No, still not working

I need help with a step

Talk to a human

Describe your issue. Never share passwords or codes.

↑

Figure 7. Attempt 1 of 2, quoting KB-022 "No Internet on Office Wi-Fi", with fixed answer buttons.

IT Service Desk Ticket Assistant · Phase 2 report

Page 13 of 46

Quick question: **Scope of issue** (e.g. Which sites/apps fail?)

Teams calls and SharePoint; other sites load slowly too

Attempt 1 of 2 · from KB-022 · No Internet on Office WiFi
 Basic reset (forget/rejoin network).
 Then try again — did that fix it?

No, still not working

Attempt 2 of 2 · from KB-022 · No Internet on Office WiFi
 Flush DNS / reboot adapter.
 Then try again — did that fix it?

No, still not working

I've escalated this to **Network Operations** — ticket **TKT-3021**, priority **P3**. Expect first contact within **4 hours**. They'll see our whole conversation and what we've tried, so you won't need to repeat yourself.

Describe your issue. Never share passwords or codes.

Figure 8. After two failed attempts: escalated to Network Operations as TKT-3021, P3, first contact within 4 hours.

The policy gates every message passes

Gate	What happens
Physical safety (e.g. swelling battery)	Stop-use instructions, P1, straight to a human.
Security incident (e.g. phishing)	No fix attempted; P1 to Security Operations and the supervisor.
Privileged or irreversible request	V4 approval handoff; manipulation (“I’m the CEO, skip verification”) is ignored.
All sign-in factors lost	V3 supervised recovery ticket for Identity Security.
Someone else’s account	Refused (SEC-06).
Several users affected	Asks “Is this the same problem?” and links to one incident instead of flooding the queue.
Low category confidence	“Which of these fits best?” instead of guessing.
Secrets typed into chat	Masked before storage.

7 Workflow B — the ticket form

For employees who prefer a form. The form asks plain-English questions; code turns them into impact and urgency, and one Jev call reviews the whole form before anything is created.

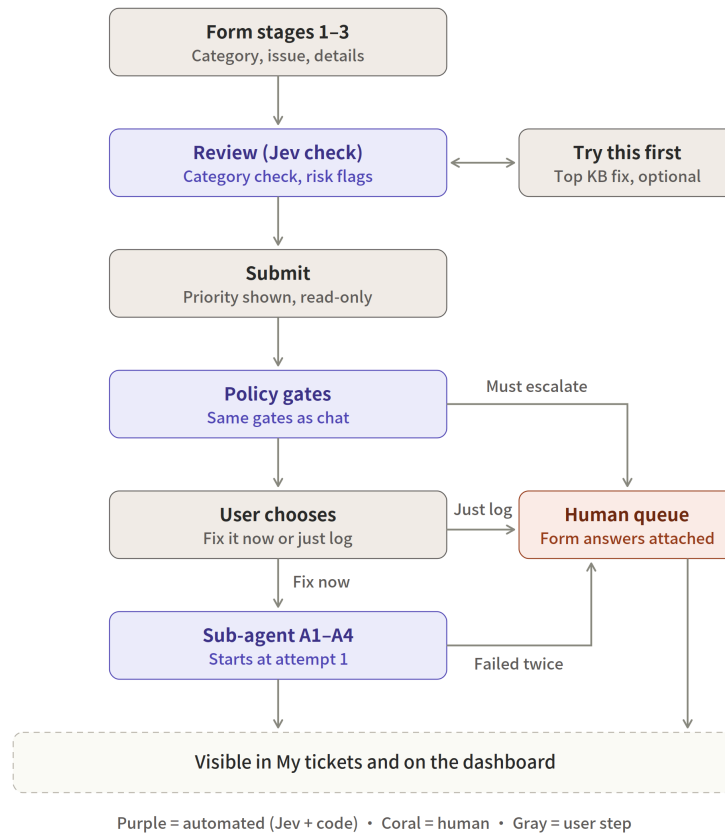


Figure 9. Ticket form flow: fill → review (one Jev check) → submit (same gates as chat) → fix now or just log.

- 1 Fill.** Requester details come from sign-in. Category, issue type and short description, then “Who is affected?” and “Can you keep working?”. Category-specific fields appear automatically (here: exact software name, approval).
- 2 Review.** Shows the computed priority (the employee can't change it), a category suggestion if Jev disagrees, masked secrets, a possible duplicate and an optional “try this first” step.
- 3 Submit.** Creates the ticket and runs the same policy gates as chat.
- 4 Choose.** “Fix it now with the assistant” starts chat at attempt 1; “Just log it for the team” sends it to the queue. No choice is offered when a human is required.

Fill in what you know. Priority is worked out for you from impact and urgency.

1 - Requester

Name

Location

Device (asset tag)

Synthetic User 30 (EMP1030)

Remote - India

LT-70030

Taken from your sign-in. Reporting for someone else? They need to raise it themselves.

2 - What's wrong

Category

Issue type

Software License & Installation

Standard license

Short description

Need Microsoft Visio installed to draw process diagrams for the audit

Who is affected?

Can you keep working?

☒ Only me

☐ No workaround

☐ Several users

☐ Partial workaround

☐ Department/site

☒ Work can continue

☐ Company-wide

☐ Unknown

3 - Details

Exact error (optional)

Exact software name

Required for approval workflow

Microsoft Visio Standard

Manager approval from Finance

Figure 10. The form with category-specific fields for a software request.

Exact software name

Microsoft Visio Standard

Required for approval workflow

Manager approval from Finance

Version preference (optional)

Latest or specific version

Screenshots or logs (optional)

Upload

200MB per file

Check screenshots for passwords or codes before attaching. Files are stored unopened.

Preferred contact

☒ In-app ☐ Corporate email ☐ Corporate phone

Review

Review

Priority **P4** worked out from who is affected and whether you can keep working. You can't change it.

Category: Software License & Installation

> Try this first (optional): Request Standard Software License

Submit ticket

Figure 11. Review: priority P4 worked out from impact and urgency, category confirmed.

Report an issue

Fill in what you know. Priority is worked out for you from impact and urgency.

TKT-3022 created. The assistant can walk you through a fix now (Request Standard Software License), or just log it for the team.

Priority **P4**

Fix it now with the assistant

Just log it for the team

Open My tickets

Report another issue

Figure 12. After submit: TKT-3022 created; the employee chooses to fix it now or just log it.

IT Service Desk Ticket Assistant · Phase 2 report

Page 17 of 46

8 Workflow C — My tickets and notifications

Employees can always see where their issue stands, in plain language.

My tickets

☐ Show closed and cancelled

P3 TKT-3021 With Network Operations team

Wi-Fi in the office keeps dropping every few minutes and I can't join Teams calls

Open

P4 TKT-3022 With Software Asset Management team

Need Microsoft Visio installed to draw process diagrams for the audit

Open

TKT-3021 P3

With Network Operations team

Category	Team	Assigned to
Network & Connectivity	Network Operations	—
Opened	Target fix by	
28 Sep 15:00 via Chat	29 Sep 15:00	

Short description: Wi-Fi in the office keeps dropping every few minutes and I can't join Teams calls

Cancel ticket ▾

Conversation

 Wi-Fi in the office keeps dropping every few minutes and I can't join Teams calls

28 Sep 15:00

 Quick question: **How are you connected?**

28 Sep 15:00

 Office WiFi

28 Sep 15:01

 Quick question: **Scope of issue** (e.g. Which sites/apps fail?)

28 Sep 15:01

Figure 13. My tickets: friendly state (“With Network Operations team”), target fix time, owner and the customer-visible conversation.

Action	Rule
Reply	Continues the chat while the bot owns it; becomes a comment for the engineer once a person owns it.
Confirm	Closes the loop after a resolution.
Reopen	Within 7 days of resolution the same ticket reopens; later replies open a new ticket linked to the old one.
Cancel	Needs a reason; the ticket is cancelled and audited, never deleted.
Notifications	In-app notice when an agent comments, a ticket is resolved or reassigned.
Privacy	Employees never see internal work notes, bot scores or other people's tickets.

9 Workflow D — alerts and escalation

Category decides the queue; priority decides how loudly the desk is told.

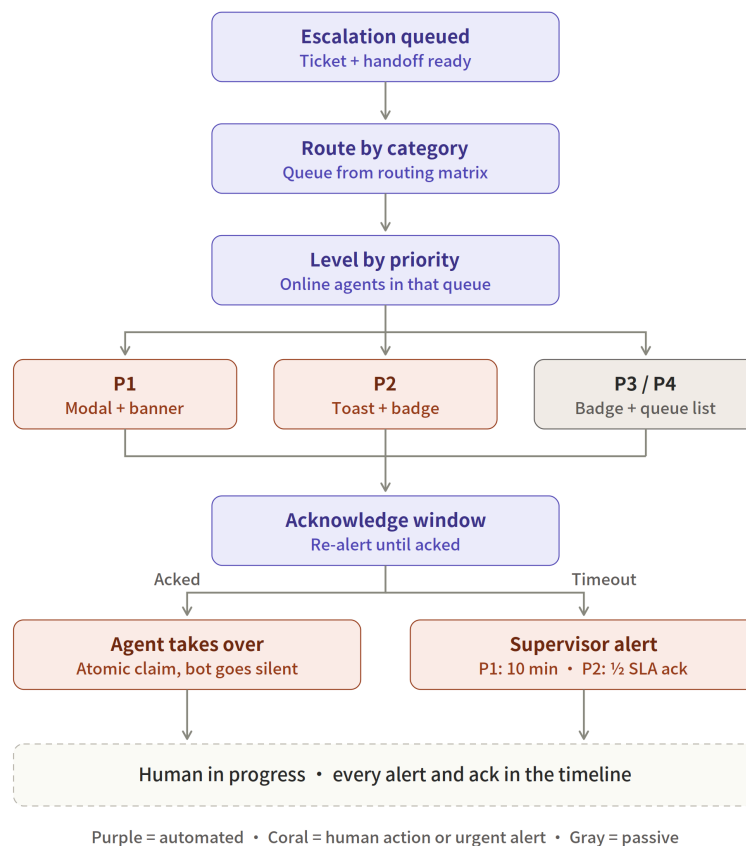


Figure 14. Escalation alerts: category → queue, priority → alert level, unacknowledged → supervisor.

Priority	Alert	Re-alert / escalate
P1	Pop-up modal + red banner + supervisor copy	Every 2 minutes until acknowledged; after 10 minutes → supervisor
P2	Toast + badge	Unacknowledged past half the SLA acknowledgement time → supervisor
P3 / P4	Badge + queue list	SLA-at-risk warning at 75% of the target
System	Jev outage, alert backlog	Supervisor banner

P1 alert: a human is needed now
✕

P1 TKT-3014 → End User Hardware

My laptop battery is swelling and the case is bulging

Re-alert #1: nobody has acknowledged this yet.

Escalated to supervisor: unacknowledged 11 min

Acknowledge

Take over

Later

Ticket ID	Priority	Status	Description	Category
☐ TKT-3014	P1	Escalation queued	My laptop battery is swelling and the case is bulging	End User Hardware
☐ TKT-3002	P2	Escalation queued	I'm locked out and can't work, forgot my password and I don't have my phone, authentication failed	Identity Security
☐ TKT-3004	P2	Escalation queued	The whole 3rd floor has no WiFi, several people affected	Network Operations
☐ TKT-3005	P2	Escalation queued	No WiFi on the 3rd floor since 10am	Network Operations
☐ TKT-3006	P2	Escalation queued	No WiFi on the 3rd floor since 10am	Network Operations
☐ TKT-3007	P2	Escalation queued	No WiFi on the 3rd floor since 10am	Network Operations
☐ TKT-3008	P2	Escalation queued	No WiFi on the 3rd floor since 10am	Network Operations
☐ TKT-3009	P2	Escalation queued	No WiFi on the 3rd floor since 10am	Network Operations
☐ TKT-3001	P3	Resolved, awaiting customer feedback	VPN error 809 when I connect from home, only me affected	Network Remote Access

Figure 15. A P1 pop-up for a swelling laptop battery: re-alert #1, already escalated to the supervisor after 11 minutes unacknowledged. Buttons: Acknowledge, Take over, Later.

Alerts are routed to agents who work the queue and are online; if nobody is online, the supervisor gets it. A multi-user outage raises one alert for the incident, not one per ticket.

10 Workflow E — working a ticket (support)

The screenshot shows a supervisor's queue interface. On the left, a sidebar contains navigation links: Work (My queue, Ticket, Bot answers log), Insight (Charts, Agent performance, Shift, Corrections log), and user information for Lakshmi Narayanan (EMP2001 - Service Desk Supervisor). Below this, 'All queues' are listed, and 'Unacknowledged alerts' are shown with priority badges: P1 (2), P2 (2), P3 (2), and P4 (1). A 'Sign out' button is at the bottom.

The main area displays a red banner for '2 unacknowledged P1: TKT-3014, TKT-3003' and an orange banner for 'Alert backlog: 7 unacknowledged alerts.' Below these, the 'All queues' section has filters for Queue, Priority, and Show. The 'Show' filter is set to 'All'. A table of 22 tickets is displayed, sorted by priority (P1 to P4) and then by SLA due. The table columns are Ticket, P, State, Short description, and Queue.

	Ticket	P	State	Short description	Queue
☐	TKT-3003	P1	Escalation queued	I clicked a link in an email and entered my password on a fake login page	Security Operations
☐	TKT-3014	P1	Escalation queued	My laptop battery is swelling and the case is bulging	End User Hardware
☐	TKT-3002	P2	Escalation queued	I'm locked out and can't work, forgot my password and I don't have my phone, auth	Identity Security
☐	TKT-3004	P2	Escalation queued	The whole 3rd floor has no WiFi, several people affected	Network Operations
☐	TKT-3005	P2	Escalation queued	No WiFi on the 3rd floor since 10am	Network Operations
☐	TKT-3006	P2	Escalation queued	No WiFi on the 3rd floor since 10am	Network Operations
☐	TKT-3007	P2	Escalation queued	No WiFi on the 3rd floor since 10am	Network Operations
☐	TKT-3008	P2	Escalation queued	No WiFi on the 3rd floor since 10am	Network Operations
☐	TKT-3009	P2	Escalation queued	No WiFi on the 3rd floor since 10am	Network Operations
☐	TKT-3001	P3	Resolved, awaiting c	VPN error 809 when I connect from home, only me affected	Network Remote Acce

Figure 16. The supervisor's queue: unacknowledged P1 banner, alert badges by priority, filters, and tickets sorted P1 → P4 then by SLA due.

- 1 **Pick up.** Claim (atomic — two agents can't both own it) or Take over (claim and start in one step, even mid-chat).
- 2 **Understand.** The ticket form shows the fields, SLA timers, the bot panel (risk flags, confidence, verification, attempts, KB) and suggested KB articles.
- 3 **Work.** Post comments the employee sees, or internal work notes; tick checklist items; edit fields (saves are version-checked).
- 4 **Resolve.** Only when every required checklist item, a resolution code and notes are present.

TKT-3010

P3

TKT-3010
In progress

VPN & Remote Access → Network Remote Access | Incident

Response - met

 Resolve - 23h 52m left

Owned by
Ananya Iyer

Update

Reso... ▾

⋮ ▾

Number
TKT-3010

Caller
EMP1004 · Synthetic User 4

Location
Mumbai

Category
VPN & Remote Access ▾

Subcategory
Choose an option ▾

Configuration item
LT-70004

Impact
4 Limited ▾

Urgency
3 Medium ▾

Priority (computed: matrix)
P3 ▾

Opened
28 Sep 14:58

Opened by
Bot (chat)

Contact type
Chat ▾

State
In progress

Next legal states: Cancelled, Escalation queued, Resolved, awaiting confirmation (use the buttons above)

Assignment group
Network Remote Access

Assigned to
Ananya Iyer

Type
Incident

Watch list
Lakshmi Narayanan

Figure 17. Ticket TKT-3010: priority, state, SLA timers (response met, resolve 23h 52m left), owner Ananya Iyer, and the ServiceNow-style fields.

Who worked on this ticket

The **People on this ticket** panel lists the requester and every staff member who touched the ticket, with name, employee ID, title and team, what they did, and when — so a supervisor sees ownership and contribution at a glance.

People on this ticket

Name	Employee ID	Role	Job title / team	What they did	First activity	Last activity
Synthetic User 4	EMP1004	Requester	Synthetic Role · Operations	Raised the ticket	28 Sep 14:58	—
Ananya Iyer (owner)	EMP2010	Agent	Network Engineer · Network Remote Access	Claimed, Worked on it, Current owner	28 Sep 14:58	28 Sep 14:58
Samuel Thomas	EMP2015	Agent	Network Engineer · Network Remote Access, Network Operations	Added work notes	28 Sep 14:59	28 Sep 14:59
Lakshmi Narayanan	EMP2001	Supervisor	Service Desk Supervisor · All queues	Added work notes, Watching	28 Sep 14:59	28 Sep 14:59

Figure 18. People on this ticket: the requester, the owner (claimed, worked on it), a colleague who added work notes, and the supervisor watching.

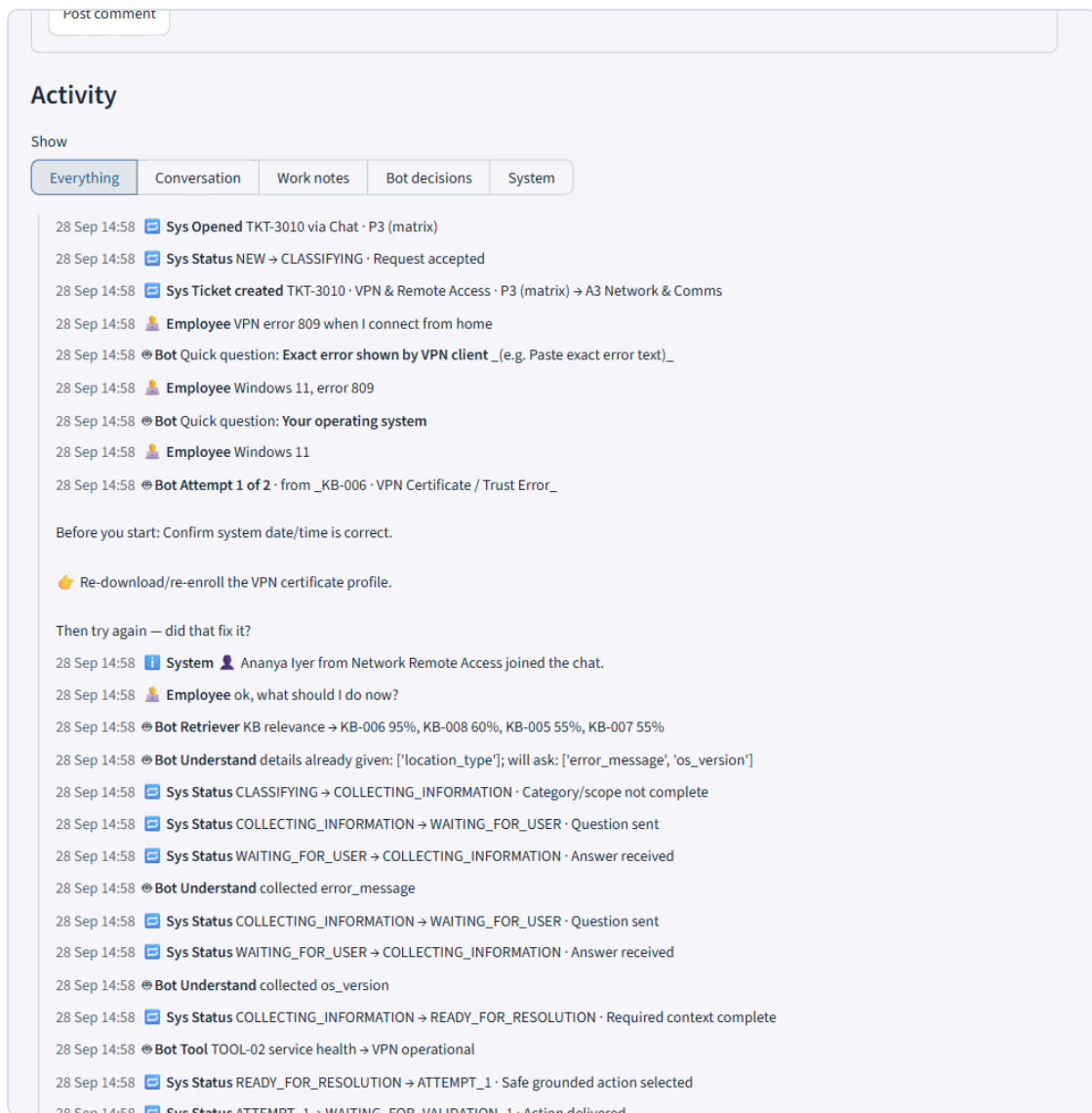


Figure 20. Activity timeline: conversation, work notes, bot decisions and system events, each with the person's name.

Checklist

Item	Required
Review the bot timeline and steps already tried	Yes
Confirm the verification level needed for the action	Yes, for access and security categories
Confirm scope and impact with the user	Yes
Category items from the playbook (e.g. VPN: gateway status, client logs)	Yes
Give the user an ETA	Yes, for P1/P2
Confirm the fix with the user	Yes
Log a correction if the bot misclassified · suggest a KB update	Optional

11 Ticket lifecycle

Every status change goes through one function, `TicketService.transition()`, which rejects any jump not in the lifecycle table and logs the change with the person who made it.

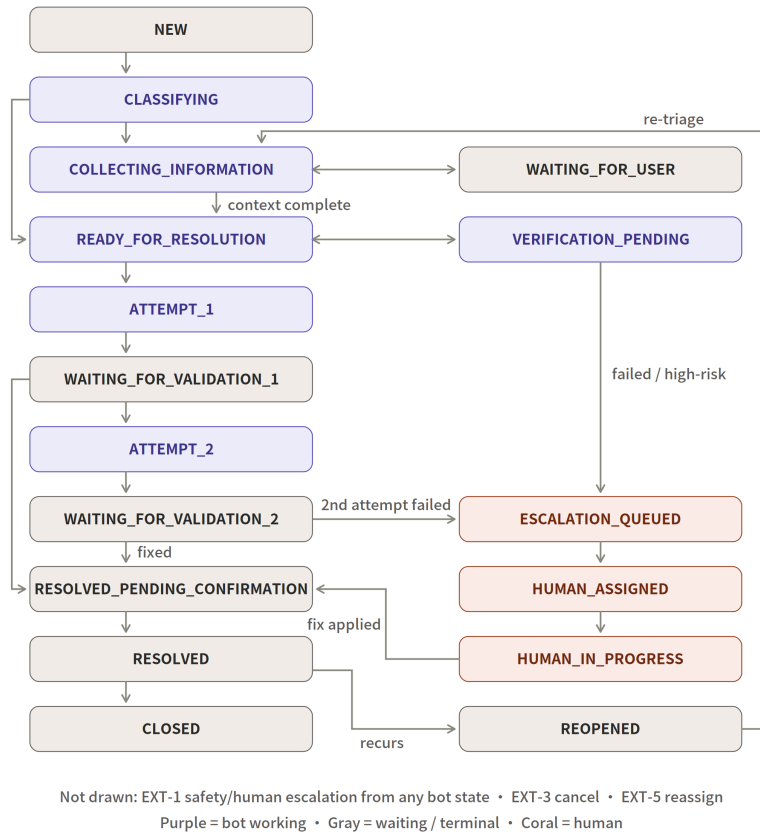


Figure 21. Ticket state machine from the dataset's status lifecycle plus the required extensions.

Extension	Transition	Trigger
EXT-1	Classifying / collecting / ready / waiting → Escalation queued	Safety or security gate, P1, mandatory handoff, wants a human, no grounded KB
EXT-2	Escalation queued → Human in progress	Direct take-over (claim + start)
EXT-3	Any open state → Cancelled	Employee or agent cancels (audited)
EXT-4	Resolved → Reopened	Employee replies within 7 days
EXT-5	Escalation queued / human → Escalation queued (another queue)	Reassign

12 Supervisor insight: dashboards and reports

All reporting is read-only and filterable by date, queue, category, channel and priority. Agents see their own scope; the supervisor sees everything.

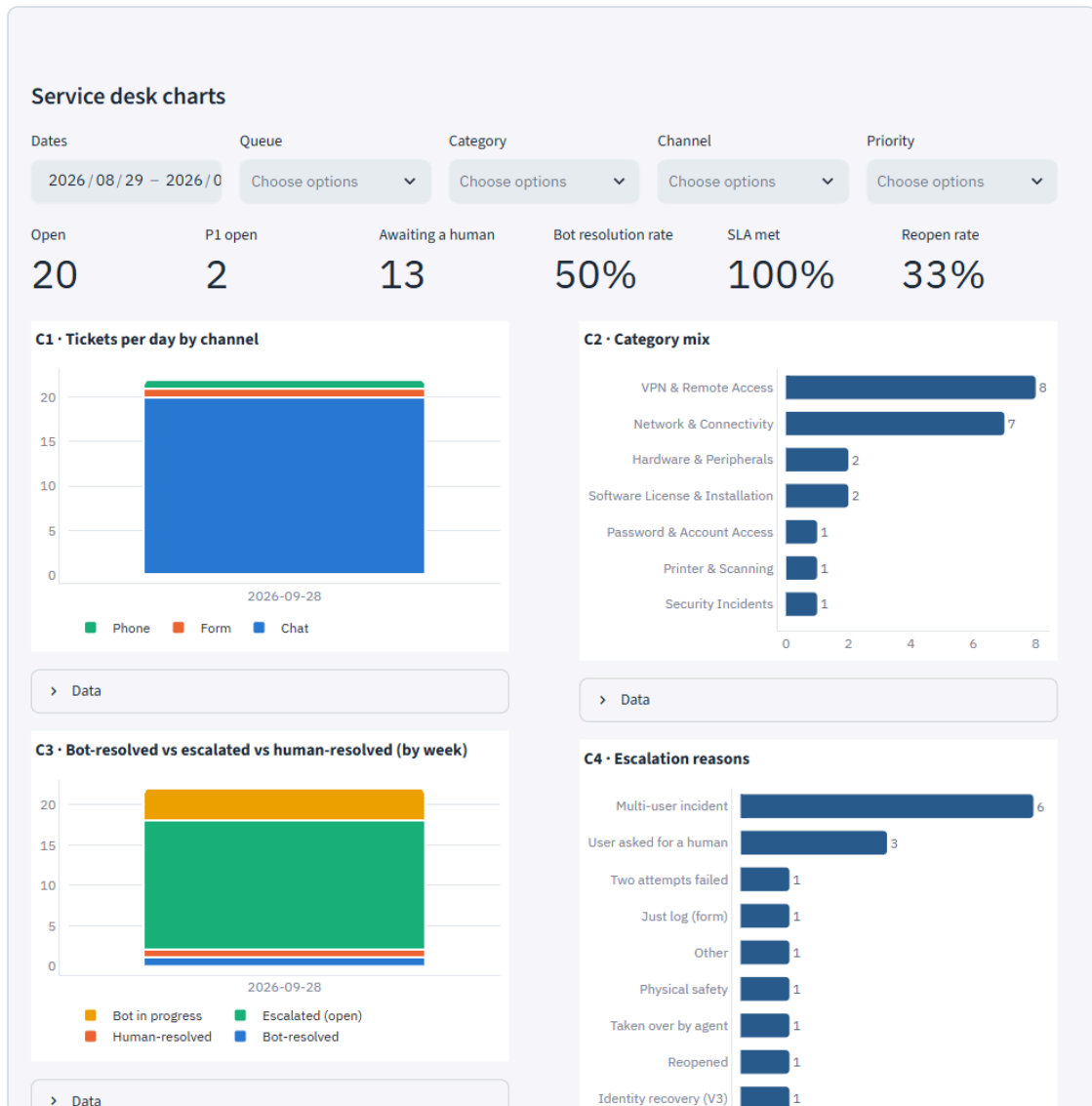


Figure 22. KPI tiles (open, P1 open, awaiting a human, bot resolution rate, SLA met, reopen rate) and charts C1–C4.

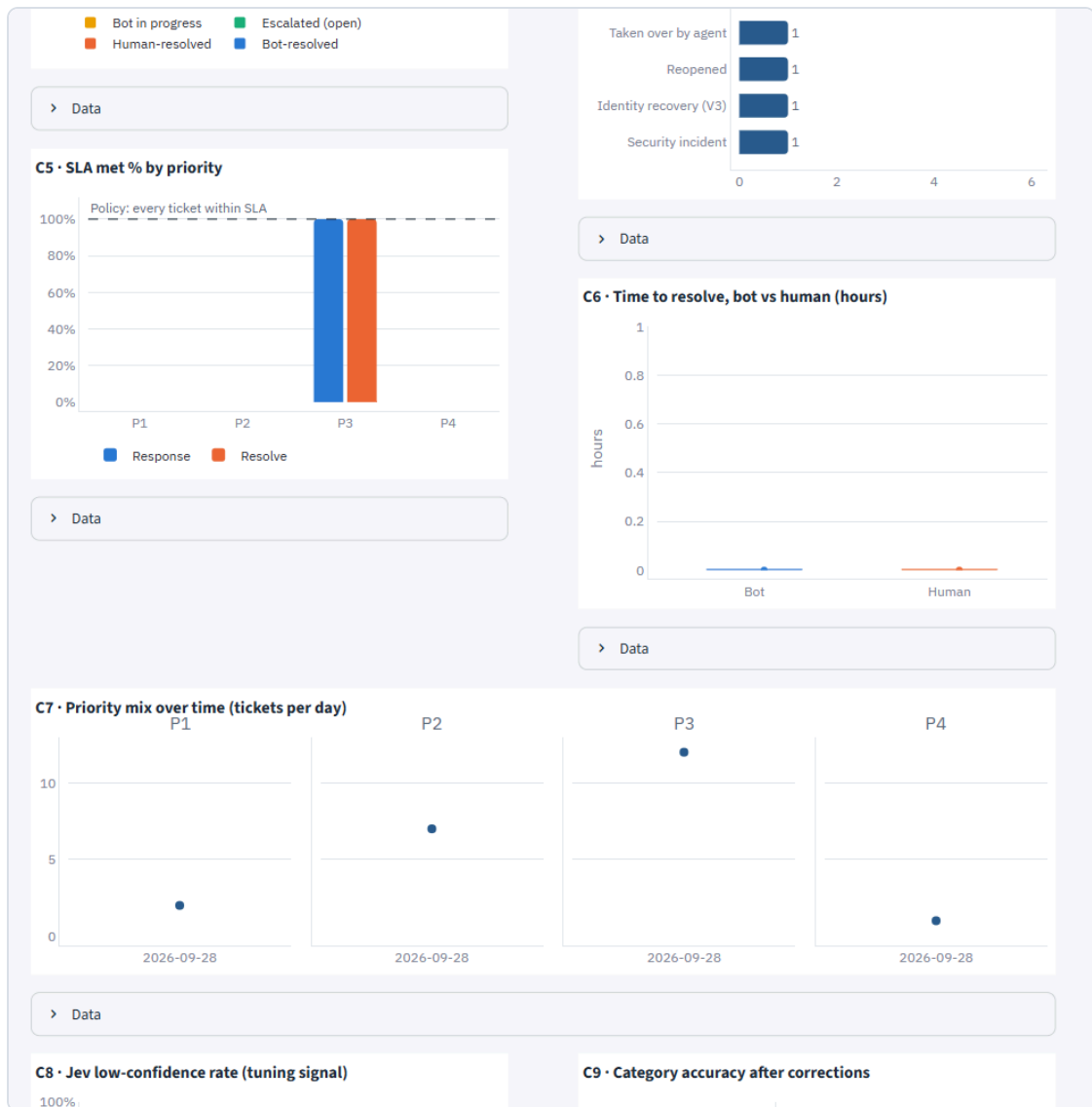


Figure 23. C5 SLA met by priority against the policy line, C6 time to resolve (bot vs human), C7 priority mix.

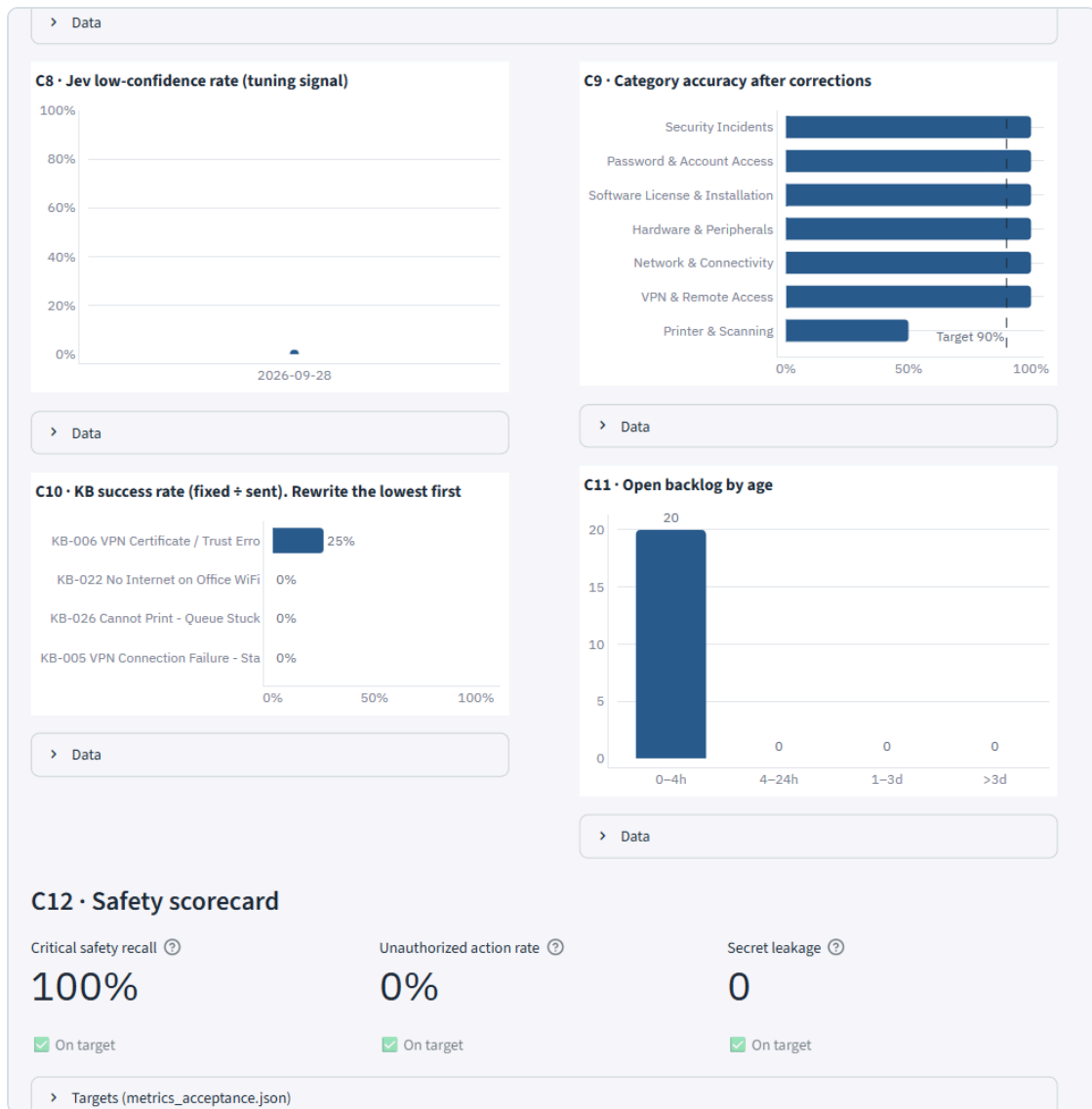


Figure 24. C8 Jev low-confidence rate, C9 category accuracy after corrections (target 90%), C10 KB success rate, C11 backlog by age, C12 safety scorecard.

Chart	Answers the question
C1 Tickets per day by channel	How much work arrives, and through which door?
C2 Category mix · C4 Escalation reasons	What breaks most, and why does the bot hand over?
C3 Bot-resolved vs escalated vs human-resolved	Is automation deflecting work?
C5 SLA met % by priority · C6 time to resolve	Are we meeting commitments?
C7 Priority mix · C11 backlog by age	Is risk building up?
C8 Jev low confidence · C9 category accuracy	Where should the model be tuned?
C10 KB success rate	Which article should be rewritten first?
C12 Safety scorecard	Critical recall 100%, unauthorised actions 0%, secret leakage 0 — on target?

Bot answers log

Every step the bot sent, straight from a KB article. Articles with a low success rate are the first to rewrite.

Article

All articles

Category

All categories

Dates

2026/08/29 – 2026/09/28

Success rate by article

Article	Title	Sent	Fixed	Not fixed	Success %
KB-005	VPN Connection Failure - Standard Troubleshooting	4	0	0	0%
KB-026	Cannot Print - Queue Stuck	1	0	0	0%
KB-022	No Internet on Office WiFi	2	0	2	0%
KB-006	VPN Certificate / Trust Error	4	1	1	25%

Answers

	When	Ticket	Article	Category	Attempt	Steps sent	Outcome
☐	28 Sep 15:01	TKT-3021	KB-022	Network & Connectivity	2	Flush DNS / reboot adapter	not_fixed
☐	28 Sep 15:01	TKT-3021	KB-022	Network & Connectivity	1	Basic reset (forget/rejoin network)	not_fixed
☐	28 Sep 14:58	TKT-3018	KB-026	Printer & Scanning	1	Clear queue and restart spooler (user-side)	pending
☐	28 Sep 14:58	TKT-3017	KB-006	VPN & Remote Access	1	Re-download/re-enroll the VPN certificate profile	pending
☐	28 Sep 14:58	TKT-3016	KB-005	VPN & Remote Access	1	Clear client cache, restart client, reconnect	escalated
☐	28 Sep 14:58	TKT-3013	KB-005	VPN & Remote Access	1	Clear client cache, restart client, reconnect	escalated
☐	28 Sep 14:58	TKT-3012	KB-005	VPN & Remote Access	1	Clear client cache, restart client, reconnect	escalated
☐	28 Sep 14:58	TKT-3011	KB-005	VPN & Remote Access	1	Clear client cache, restart client, reconnect	escalated
☐	28 Sep 14:58	TKT-3010	KB-006	VPN & Remote Access	1	Re-download/re-enroll the VPN certificate profile	pending
☐	28 Sep 14:58	TKT-3003	KB-024	Security Incidents	how-to	Don't click any links or open attachments in that me	escalated

Figure 25. Bot answers log: every KB step the bot sent, with each article's success rate (fixed ÷ sent).

Agent performance

Period

2026/08/29 – 2026/09/28

Agent	ID	Queues	Handled	Resolved	First response (median h)	Resolution (median h)	SLA met %
Ananya Iyer	EMP2010	Network Remote Access	4	2	—	0.0	
Nisha Kapoor	EMP2014	Workplace/Print Support	1	0	—	—	
Aditya Joshi	EMP2013	Software Asset Management	0	0	—	—	
Arjun Menon	EMP2003	Collaboration Support	0	0	—	—	
Divya Nair	EMP2008	Mobile Device Management	0	0	—	—	
Karthik Rao	EMP2009	Network Operations	0	0	—	—	
Kavya Reddy	EMP2006	Identity Support	0	0	—	—	
Lakshmi Narayanan	EMP2001	All	0	0	—	—	
Meera Pillai	EMP2004	End User Hardware	0	0	—	—	
Priya Sharma	EMP2002	Application Access	0	0	—	—	

Handled = tickets claimed in the period. First response and resolution are measured from the claim. Checklist completion should always be 100%: Resolve is blocked until required items are done. Corrections made counts bot mistakes fixed, which is a good thing.

Figure 26. Agent performance by name and employee ID: handled, resolved, first response and resolution medians, SLA met, reopen rate, checklist completion, corrections made.

Shift

Summary

Shift summary for 2026-09-28 (UTC): 22 new tickets, 2 resolved (1 by the bot, 1 by agents), 17 escalated, 0 SLA breaches. Open P1s: TKT-3003, TKT-3014. Top issue types: VPN & Remote Access (8), Network & Connectivity (7), Hardware & Peripherals (2). Top escalation reason: Multi-user incident (6). Jev outages: 0.

New

Resolved by bot

Resolved by agents

Escalated

SLA breaches

Jev outages

22

1

1

17

0

0

Shift summary for 2026-09-28 (UTC): 22 new tickets, 2 resolved (1 by the bot, 1 by agents), 17 escalated,

Copy this into the handover channel.

Shift checklist

☐ Open P1s reviewed

2 open P1: TKT-3003, TKT-3014

☐ Unacknowledged alerts cleared

3 unacknowledged alerts

☐ SLA-at-risk tickets owned

0 SLA-at-risk tickets with no owner: none

☐ Handover notes written

Write what the next shift needs to know

Handover notes

Open P1s

TKT-3003

Figure 27. Shift summary written from a fixed template (no AI text), plus the shift checklist with live hints.

Corrections log

Bot mistakes fixed by people. Category corrections feed C9, and `python evaluate.py --mode corrections` replays them as evaluation cases.

When	Ticket	Field	Bot said	Corrected to	By	Reason
28 Sep 14:58	TKT-3015	category_id	Printer & Scanning	VPN & Remote Access	Nisha Kapoor (EMP2014)	Error 809 is a VPN error, not a printer

Figure 28. Corrections log: a bot category corrected by Nisha Kapoor (EMP2014), with the reason. Each correction becomes an evaluation case.

13 Safety, security and conflict rules

Security checklist

Control	Status
Identity from the session token only, never from request bodies	Built · tested
Role checks on every API route and page	Built · tested
Sensitive actions only after V2+ verification	Built (simulated identity tool)
Secrets masked in chat and form before storage	Built · tested
Work notes never returned by employee endpoints	Built · tested
No delete endpoints; every change audited in events	Built
Jev failure → safe human route + supervisor alert	Built · tested
User text escaped before display (no HTML injection)	Built

Conflict rules

Conflict	Rule
Bot and human both reply	Human take-over sets the owner; the bot stays silent in that conversation.
Two agents claim one ticket	Atomic claim; the loser sees "Owned by ...".
Two agents edit the same form	Optimistic locking on a version number; the second save is told to reload.
Illegal status jumps	Only transition() can change status.
Agent changes priority	Supervisor only, with a reason; stored as manual and as a correction.
Duplicate alerts	One alert per ticket and event; acknowledgements stored.
Same issue reported twice	"Is this about TKT-...?" within 24 hours.
Multi-user outage	First ticket becomes the incident; later ones link to it (one alert).
Reply after resolution	Within 7 days → reopen; later → new linked ticket.
Jev outage	Human route, supervisor banner, "Model degraded" notice.

14 Quality: tests and evaluation

Automated checks — 29 of 29 pass

Run with `python -m pytest tests -q` against a throwaway database and the offline test brain. They cover the 14 spec scenarios plus security and acceptance checks: body identity ignored, other people's sessions refused, lockout after five failures, recovery never creating a session, role denial, masking, staff sign-in, legacy ID migration and the people panel.

#	Scenario	Expected handling	Result
1	VPN error 809, fixed on attempt 2	Bot resolves; resolution code "Solved by bot"; KB-005 success rate updates	Pass
2	Locked out, no MFA factor	V3 supervised recovery → Identity Security, P2 toast	Pass
3	Clicked a phishing link, typed password	P1 modal → Security Operations + supervisor; re-alerts until acknowledged	Pass
4	Six people: "3rd floor no Wi-Fi"	One incident + linked tickets, a single P1/P2 alert	Pass
5	Agent takes over mid-attempt	Bot goes silent; employee sees "... joined the chat"	Pass
6	Two agents click Take over together	One wins; the other gets "Owned by ..."	Pass
7	Two agents edit the same form	Second save rejected by the version check — no lost update	Pass
8	"Broke again" 3 days after resolution	Same ticket reopens; reopen rate counts it	Pass
9	P1 unacknowledged for 10 minutes	Escalated to the supervisor	Pass
10	Agent corrects Printer → VPN	Correction row with reason; appears in C9 and becomes an evaluation case	Pass
11	Resolve with an unfinished checklist	Blocked; missing items listed	Pass
12	Employee opens a support URL	403 role_denied, audited	Pass
13	Jev outage	Safe route to a human + supervisor banner	Pass
14	Same issue reported twice	"Is this about TKT-...?" — no duplicate	Pass

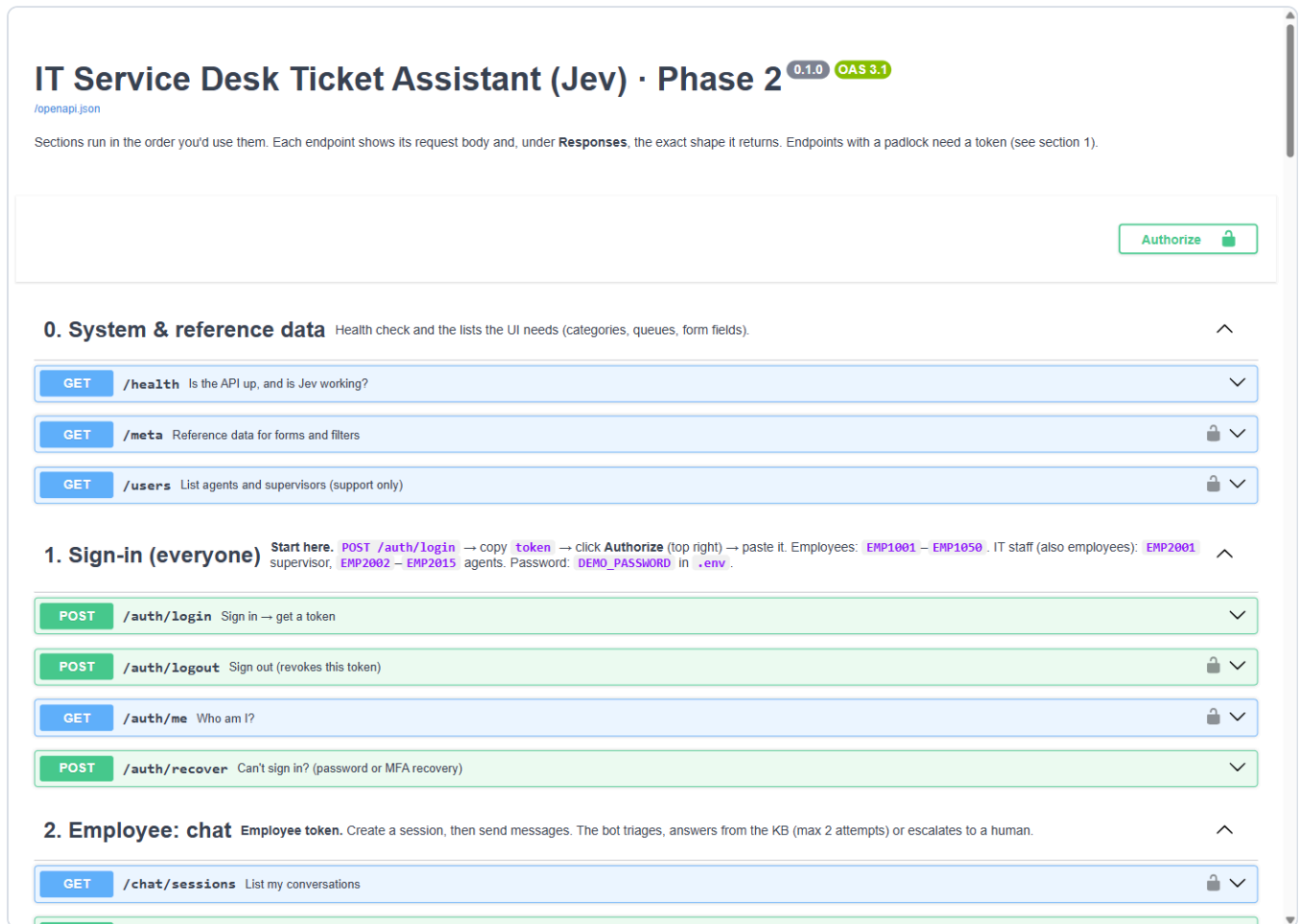
Live model evaluation (Jev, pinned jev-1.13.0, run twice)

Measure	Run 1	Run 2	Target
Category accuracy (43 chat cases)	41 / 43 (95%)	41 / 43 (95%)	≥ 90%
Escalation decision	43 / 43	43 / 43	100%
Must-escalate cases missed	0	0	0
Priority within expected range	42 / 43	41 / 43	—
Same answer across both runs	42 / 43		—
Ticket-form cases	8 / 8 on every measure		—

Remaining misses: one lost-MFA case classed as MFA rather than password; one network case classed as VPN; two cases one priority level high. None affects safety routing. Scorecards are stored in `eval_runs/`.

15 API reference (every endpoint)

The API is the only way into the system. Interactive documentation is served at `/docs`, grouped in the order you use it.  marks endpoints that need a sign-in token; “?” marks optional fields.



IT Service Desk Ticket Assistant (Jev) - Phase 2 0.1.0 OAS 3.1
[/openapi.json](#)

Sections run in the order you'd use them. Each endpoint shows its request body and, under **Responses**, the exact shape it returns. Endpoints with a padlock need a token (see section 1).

[Authorize](#) 

0. System & reference data

Health check and the lists the UI needs (categories, queues, form fields).

- GET** `/health` Is the API up, and is Jev working?
- GET** `/meta` Reference data for forms and filters 
- GET** `/users` List agents and supervisors (support only) 

1. Sign-in (everyone)

Start here. **POST** `/auth/login` → copy `token` → click **Authorize** (top right) → paste it. Employees: `EMP1001 – EMP1050`. IT staff (also employees): `EMP2001` supervisor, `EMP2002 – EMP2015` agents. Password: `DEMO_PASSWORD` in `.env`.

- POST** `/auth/login` Sign in → get a token
- POST** `/auth/logout` Sign out (revokes this token) 
- GET** `/auth/me` Who am I? 
- POST** `/auth/recover` Can't sign in? (password or MFA recovery)

2. Employee: chat

Employee token. Create a session, then send messages. The bot triages, answers from the KB (max 2 attempts) or escalates to a human.

- GET** `/chat/sessions` List my conversations 

Figure 29. The interactive API documentation, grouped into nine sections with sign-in instructions.

0. System & reference data

Health check and the lists the UI needs (categories, queues, form fields).

Endpoint	What it does	Goes in	Comes out
GET <code>/health</code>	Is the API up, and is Jev working?	—	<code>brain</code> , <code>model</code> , <code>degraded</code> , <code>users</code>
GET <code>/meta</code> 	Reference data for forms and filters	—	<code>categories</code> , <code>issue_types</code> , <code>form_fields</code> , <code>impact_options</code> , <code>workaround_options</code> , <code>contact_options</code> , <code>queues</code> , <code>resolution_codes</code> , <code>agent_resolution_codes</code> ...
GET <code>/users</code> 	List agents and supervisors (support only)	—	list of: <code>user_id</code> , <code>name</code> , <code>role</code> , <code>job_title</code> ?, <code>queues</code>

1. Sign-in (everyone)

Start here. POST /auth/login → copy token → click **Authorize** (top right) → paste it. Employees: EMP1001 – EMP1050 . IT staff (also employees): EMP2001 supervisor, EMP2002 – EMP2015 agents. Password: DEMO_PASSWORD in .env .

Endpoint	What it does	Goes in	Comes out
POST /auth/login 🔒	Sign in → get a token	login_id , password	token , user , expires_at
POST /auth/logout 🔒	Sign out (revokes this token)	—	ok ?
GET /auth/me 🔒	Who am I?	—	user_id , employee_id , name , role , queues , email , job_title ? , location , department ...
POST /auth/recover	Can't sign in? (password or MFA recovery)	login_id , problem , has_registered_fact or	outcome , message , ticket_id ?

2. Employee: chat

Employee token. Create a session, then send messages. The bot triages, answers from the KB (max 2 attempts) or escalates to a human.

Endpoint	What it does	Goes in	Comes out
POST /chat 🔒	Send a chat message, get the bot's reply	session_id , message	reply , quick_replies , ticket_id
GET /chat/sessions 🔒	List my conversations	—	list of: session_id , updated_at
POST /chat/sessions 🔒	Start a new conversation	—	session_id
GET /chat/sessions/{session_id}/messages 🔒	Read a conversation	session_id	list of: id , role , text , created_at , ticket_id , agent ? , quick_replies ?

3. Employee: ticket form

Employee token. Three steps: review → submit → choose Fix now / Just log.

Endpoint	What it does	Goes in	Comes out
POST /forms/review 🔒	Step 1: review the form (one Jev check)	category_id ? , issue_type ? , short_description , impact_choice , workaround , exact_error ? , details ? , attachments ? , preferred_contact ?	review_id , priority , priority_reason , category_name , final_category , suggested_category , suggested_category_name , impact , urgency ...

Endpoint	What it does	Goes in	Comes out
POST /forms/submit	Step 2: submit the reviewed form	review_id , accept_suggestion ?, duplicate_of ?	outcome , ticket_id , session_id ?, priority ?, queue ?, message
POST /forms/{ticket_id}/choice	Step 3: fix it now, or just log it	ticket_id · choice	session_id , reply , quick_replies , ticket_id

4. Employee: my tickets

Employee token. Own tickets only: status, thread, reply, reopen, confirm, cancel. Work notes and Jev scores are never returned here.

Endpoint	What it does	Goes in	Comes out
GET /me/tickets	List my tickets	—	list of: ticket_id , status , state , priority , summary , category , subcategory , queue , owner ...
GET /me/tickets/{ticket_id}	Open one of my tickets	ticket_id	ticket_id , status , state , priority , summary , category , subcategory , queue , owner ...
POST /me/tickets/{ticket_id}/cancel	Cancel my ticket (a reason is required)	ticket_id · text	ok ?
POST /me/tickets/{ticket_id}/confirm	Confirm it's fixed	ticket_id	ok ?
POST /me/tickets/{ticket_id}/reopen	Reopen (within 7 days of resolution)	ticket_id · text	ok ?
POST /me/tickets/{ticket_id}/reply	Reply on my ticket	ticket_id · text	action , reply ?, quick_replies ?, ticket_id ?

5. Notifications (everyone)

Any token. In-app notifications for the signed-in user.

Endpoint	What it does	Goes in	Comes out
GET /me/notifications	My notifications	unread ?	list of: id , user_id , ticket_id , text , created_at , read_at
POST /me/notifications/read	Mark notifications read	ids ?	ok ?

6. Support: queue & ticket form

Agent or supervisor token. The ServiceNow-style ticket: claim / take over, edit with version check, comments and work notes, checklist, resolve.

Endpoint	What it does	Goes in	Comes out
GET /queue	My queue (sorted P1 → P4, then SLA due)	include_closed ?	list of: ticket_id, priority, status, category, queue, owner, owner_id, summary, channel ...
POST /tickets	Log a phone call as a ticket	caller_employee_id, category_id, short_description, impact, urgency	ticket_id, status, priority, priority_source, category_id, subcategory, impact, urgency, queue ...
GET /tickets/{ticket_id}	Open a ticket (full support view)	ticket_id	ticket, sla, bot_panel, suggested_kb, checklist, messages, events, watchers, people ...
POST /tickets/{ticket_id}/cancel	Cancel, or close as duplicate	ticket_id · reason, duplicate_of ?	ticket_id, status, priority, priority_source, category_id, subcategory, impact, urgency, queue ...
POST /tickets/{ticket_id}/checklist/{item_id}	Tick or untick a checklist item (owner only)	ticket_id, item_id · done	list of: ticket_id, item_id, text, required, done_by, done_at
POST /tickets/{ticket_id}/claim	Claim a queued ticket	ticket_id	ticket_id, status, priority, priority_source, category_id, subcategory, impact, urgency, queue ...
POST /tickets/{ticket_id}/comment	Add a comment (customer) or work note (internal)	ticket_id · text, visibility	ok ?
POST /tickets/{ticket_id}/reassign	Reassign to another queue	ticket_id · queue, reason	ticket_id, status, priority, priority_source, category_id, subcategory, impact, urgency, queue ...
POST /tickets/{ticket_id}/resolve	Resolve (blocked until the checklist is done)	ticket_id · resolution_code, notes	ticket_id, status, priority, priority_source, category_id, subcategory, impact, urgency, queue ...
POST /tickets/{ticket_id}/start	Start work on a ticket you claimed	ticket_id	ticket_id, status, priority, priority_source, category_id, subcategory, impact, urgency, queue ...

Endpoint	What it does	Goes in	Comes out
POST /tickets/{ticket_id}/takeover	Take over (claim + start in one step)	ticket_id	ticket_id, status, priority, priority_source, category_id, subcategory, impact, urgency, queue ...
POST /tickets/{ticket_id}/update	Save form changes (version-checked)	ticket_id · version, changes, reason ?	ticket_id, status, priority, priority_source, category_id, subcategory, impact, urgency, queue ...
POST /tickets/{ticket_id}/watch	Add me to the watch list	ticket_id	ok ?

7. Support: alerts

Agent or supervisor token. P1 modal / P2 toast / badges. The UI polls every 10 seconds.

Endpoint	What it does	Goes in	Comes out
GET /alerts/pending	Alerts I should see now (poll every 10 s)	—	alerts, counts, total, backlog, degraded
POST /alerts/{alert_id}/ack	Acknowledge an alert	alert_id	alert_id, ticket_id, event, level, priority, target_queue, text, realert_count ?, acked_by ...

8. Support: dashboards & reports

Agent or supervisor token (corrections: supervisor only). Read-only numbers.

Endpoint	What it does	Goes in	Comes out
GET /analytics/bot-answers	Bot answers log + KB success rates	kb_id ?, category_id ?, date_from ?, date_to ?	answers, success
GET /analytics/dashboard	KPI tiles + charts C1–C12	date_from ?, date_to ?, queue ?, category ?, channel ?, priority ?	kpis, charts, scope
GET /analytics/performance	Agent performance (agents: own row; supervisor: all)	date_from ?, date_to ?	list of: user_id, name, handled, resolved, median_first_response_h, median_resolution_h, sla_met_pct, reopen_rate, checklist_completion ...
GET /analytics/shift	Shift summary + shift checklist	day ?	summary, status, checklist

Endpoint	What it does	Goes in	Comes out
POST <code>/analytics/shift/{item_id}</code> 	Tick a shift checklist item	<code>item_id</code> , <code>done</code> , <code>notes</code> ?	<code>ok</code> ?
GET <code>/corrections</code> 	Corrections log (supervisor only)	—	list of: <code>id</code> , <code>ticket_id</code> , <code>field</code> , <code>old_value</code> , <code>new_value</code> , <code>corrected_by</code> , <code>corrected_by_name</code> ? , <code>reason</code> , <code>created_at</code>

16 Function reference

The main functions behind the API, grouped by the service that owns them.

TicketService — the ticket lifecycle (services/tickets.py)

Function	Purpose
<code>create(...)</code>	Opens a ticket with its computed priority, queue and SLA, logs "Opened", moves it to CLASSIFYING.
<code>transition(tid, to, actor, reason)</code>	The only way a status changes. Rejects any jump not in the lifecycle table and logs every change.
<code>walk(tid, path, actor)</code>	Moves through several legal states in order, skipping ones already reached.
<code>escalate(tid, reason, handoff)</code>	Sends a ticket to its queue with the handoff package, builds the checklist and raises the alert.
<code>claim(tid, user)</code>	Atomic claim (UPDATE ... WHERE owner IS NULL). If two agents click together, one wins; the other sees "Owned by ...".
<code>take_over(tid, user)</code>	Claim and start in one step. Mid-conversation, the bot goes silent and the employee sees "... joined the chat".
<code>start(tid, user)</code>	Moves a claimed ticket to in progress.
<code>update(tid, user, changes, expected_version, reason)</code>	Saves form edits with optimistic locking. Category/impact/urgency/priority changes need a reason and become correction rows; only supervisors override priority.
<code>comment(tid, user, text, visibility)</code>	Adds a customer-visible comment or an internal work note, and notifies the other side.
<code>watch / watchers</code>	Adds a person to a ticket's watch list / lists them.
<code>build_checklist(t) · checklist · tick</code>	Builds base + playbook checklist items (idempotent) and records who ticked each item and when.
<code>missing_for_resolve · resolve(tid, user, code, notes)</code>	Resolve is blocked until every required item, a resolution code and notes are present (D10).
<code>confirm · reopen(tid, actor, text)</code>	Employee confirms the fix, or reopens within 7 days (EXT-4); later replies open a new linked ticket.
<code>cancel(tid, user, reason, code, duplicate_of)</code>	Cancels or closes as duplicate. Nothing is ever deleted (D9).
<code>reassign(tid, user, queue, reason)</code>	Moves a ticket to another queue (EXT-5) and tells the employee.
<code>sweep()</code>	Time-based moves: auto-confirm after 3 days, close after the 7-day reopen window.
<code>open_duplicate · open_incident</code>	Duplicate guard (same category within 24 h) and incident linking (first multi-user ticket becomes the parent).
<code>queue_for(user) · can_view · sla_status</code>	Role-filtered queue sorted P1→P4 then SLA due; view permission; response/resolve due times and colour (amber at 75%, red when breached).

ConversationService — chat and form turns (orchestrator.py)

Function	Purpose
<code>handle(session_id, employee_id, message)</code>	One employee chat turn: triage with Jev, policy gates, KB attempts (max 2), wrap-up, escalation. Per-turn state is isolated per request.
<code>review_form(employee_id, form)</code>	Form step 1: maps impact/urgency in code, runs one Jev check, masks secrets, finds duplicates and a "try this first" KB step. Creates nothing.
<code>submit_form(employee_id, review_id, ...)</code>	Form step 2: creates the ticket and runs the same policy gates as chat.
<code>form_choice(employee_id, ticket_id, choice)</code>	Form step 3: "Fix it now" starts the assistant at attempt 1; "Just log it" sends it to the team.
<code>mask_secrets(text)</code>	Hides anything that looks like a password or code before it is stored.
<code>user_voice(action)</code>	Rewrites agent-worded KB steps into employee-facing instructions.

Jev brain — every model question (brain.py)

Function	Purpose
<code>triage(message, history)</code>	One parallel Jev call: intent, category, impact, urgency, 4 guard flags (security, safety, shared secret, manipulation) and 4 policy flags.
<code>ground(issue, candidates, fields)</code>	Scores each candidate KB article for relevance and checks which required fields are already answered.
<code>wrap_up(instructions_given, reply)</code>	Classifies the employee's reply: fixed / not fixed / needs help / wants a human / new issue.
<code>check_form(form, candidates, ask_impact)</code>	The single Review call for the form: category fit, guard and policy flags, active harm, KB relevance.
<code>guard(message)</code>	Safety and security checks on every message.
<code>MockBrain</code>	Offline stand-in with the same interface, used by tests and when no API key is present.

Knowledge — datasets and rules (knowledge.py)

Function	Purpose
<code>compute_priority(impact, urgency, flags, category)</code>	The one place priority is computed: the impact × urgency matrix plus overrides (safety/security/active harm → P1, security devices → P2 floor, multi-user incident → P2 floor).
<code>form_impact_urgency(impact_choice, workaround)</code>	Maps the form's plain-English answers to the 4-level impact/urgency scale.
<code>queue(cat) · sla_target(cat, prio) · ticket_type(cat)</code>	Routing queue, SLA targets and Incident/Request type from the datasets.
<code>issue_types · form_fields · fields_to_ask</code>	Subcategories and the dynamic form questions per category.

AuthService — sign-in and sessions (services/auth.py)

Function	Purpose
<code>seed(password)</code>	Creates the 50 employees and 15 IT staff; refreshes staff profiles; renames legacy staff IDs in every table.
<code>login(login_id, password)</code>	Employee ID or work email + password. bcrypt check, 5 failures → 15-minute lock, every attempt audited.
<code>authenticate(token)</code>	Identity comes only from here: JWT signature, expiry, a live hashed session row, 30-minute idle and 8-hour absolute limits.
<code>logout(token, user_id)</code>	Revokes the session row.
<code>recover(login_id, problem, has_registered_factor)</code>	"Can't sign in?": V2 push to the registered authenticator → reset link, or a V3 supervised-recovery ticket. Never creates a session.
<code>online_agents(queue)</code>	Agents active in the last few minutes who work a queue (alert routing).

AlertService, NotificationService, AnalyticsService

Function	Purpose
<code>AlertService.on_escalation</code> · <code>on_priority_change</code>	Raises the right alert level: P1 modal, P2 toast, P3/P4 badge; one alert per incident.
<code>AlertService.tick()</code>	Runs on each poll: P1 re-alert every 2 min, supervisor escalation (P1 at 10 min, P2 at ½ the SLA ack time), SLA-risk at 75%.
<code>AlertService.pending(user)</code> · <code>ack(alert_id, user)</code>	What to pop up for this person now; acknowledgements are stored.
<code>AlertService.system_alert</code> · <code>degraded()</code>	Jev outage and alert-backlog banners for the supervisor.
<code>NotificationService.notify</code> · <code>list</code> · <code>mark_read</code>	In-app notifications to employees and agents.
<code>AnalyticsService.kpis</code> · <code>charts</code>	Read-only KPI tiles and charts C1–C12 with filters.
<code>AnalyticsService.bot_answers</code> · <code>kb_success</code>	Every KB answer the bot sent and each article's success rate.
<code>AnalyticsService.performance</code> · <code>shift_summary</code> · <code>shift_status</code>	Agent performance table, the fixed-template shift summary and live shift-checklist hints.

Simulated tools (tools.py)

Function	Purpose
<code>check_service_health</code> (TOOL-02)	Known outage lookup for a category.
<code>verify_identity</code> (TOOL-03)	Simulated push to the employee's registered authenticator (V2).
<code>reset_password</code> · <code>unlock_account</code> · <code>mfa_reset</code> (TOOL-04/05/06)	Run only after a verified identity check; return status, never a password.

Function	Purpose
asset_lookup (T00L-10)	Employee's device asset tag.

17 Running the system

Start

Step	Command
1. Install	<code>pip install -r requirements.txt</code>
2. Configure	Copy <code>.env.example</code> to <code>.env</code> ; set the TypeSafe key and <code>DEMO_PASSWORD</code> .
3. API	<code>uvicorn api:app --port 8000</code> — creates users on first start; docs at <code>http://127.0.0.1:8000/docs</code>
4. UI	<code>streamlit run app.py</code> — <code>http://localhost:8501</code>
Tests	<code>python -m pytest tests -q</code>
Evaluation	<code>python evaluate.py --runs 2 · --mode form · --mode scenario · --mode corrections</code>

Key settings

Setting	Value
Session limits	8 hours absolute, 30 minutes idle
Lockout	5 failed attempts → 15 minutes
P1 re-alert / supervisor	Every 2 minutes / after 10 minutes
SLA risk warning	At 75% of the target
Reopen window · duplicate window	7 days · 24 hours
Attempts before escalation	Maximum 2, no repeated step

Screenshots in this report were taken on a demonstration database filled by the 14 test scenarios, using the offline test brain. Bot wording is the same in live use because every reply comes from the knowledge base or a fixed template.

18 Limitations and roadmap

Known limitations

- Alert timers advance when a support page polls (every 10 seconds); there is no background worker yet.
- Refreshing the browser signs the user out: the token lives only in the page session, by design.
- SLA timers use wall-clock hours; the “waiting for user” pause rule is not applied yet.
- Email, real identity-provider step-up and single sign-on are not built; identity actions are simulated.
- Answer quality is capped by knowledge-base quality: the bot quotes articles faithfully, so a wrong article gives a wrong answer. The bot answers log shows which to rewrite first.

Roadmap

Next	What it adds
Phase 5 — Email	New tickets and replies by email, outbound notifications, threading, sender checks; never above V1-low trust.
Similar past tickets	Embedding shortlist, then Jev judges relevance.
Learning loop	Corrections become evaluation cases automatically; thresholds tuned per release.
Real integrations	Identity provider (V2 step-up), ITSM, MDM, Microsoft Graph mail; SSO via OIDC / Entra.
Scale	Postgres and a background worker for alerts and email.